

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)(i)	(gravitational) force acts perpendicular to direction of motion	B1
	gravitational force provides centripetal acceleration	B1
1(b)(ii)	$(F =) GMm / x^2 = m\omega^2 x$ and $\omega = 2\pi / T$ or $GMm / x^2 = 4\pi^2 mx / T^2$	M1
	completion of algebra leading to $x^3 = GMT^2 / 4\pi^2$	A1
	clear indication that B = radius of planet and that A = mass (of planet)	B1
1(b)(iii)	gradient = $\sqrt[3]{(4\pi^2 / GA)}$	C1
	e.g. $(1280 - 360) / (12 \times 10^6) = \sqrt[3]{(4\pi^2 / [6.67 \times 10^{-11} \times A])}$	C1
	$A = 1.3 \times 10^{24} \text{ kg}$	A1
	intercept = gradient $\times B$	C1
	e.g. $360 = ((1280 - 360) \times B) / (12 \times 10^6)$ $B = 4.7 \times 10^6 \text{ m}$	A1

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